

PH.D. APPLIED QUALIFYING EXAM 2019 (TAKE HOME)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have technical questions about the two exam problems, you may email Prof. Sijian Wang at sijian.wang@stat.rutgers.edu (for problem 1) and Prof. Ying Hung at yhung@stat.rutgers.edu (for problem 2), cc'ing Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu). If you have questions regarding logistics (e.g., submission instructions given below), please email Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

- (1) Type up your answers to the two problems in two separate Word or Latex files.
- (2) Create a folder for each problem. The folder must contain the following:
 - (a) A word or pdf file containing your answers to the problem, including all tables, figures and Appendices.
 - (b) An R (or python) file containing the codes used to generate your results.
 - (c) A word or a text file that gives a summary of the codes submitted.
- (3) Place the two folders into a single folder titled **yourlastname_yourfirstname_Appliedexam2018**. Zip this folder and email it to hcrane@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 26, Friday.

Problem 1

BACKGROUND

As of 2010, 82% of the U.S. and 50.5% of the worldwide population lived in urban areas. As urban lands and population continue to increase, urban areas are sources of greenhouse gas emission, increased streamflow, nutrient export, and decreased biodiversity. However, urban forests provide a range of ecosystem services to offset these negative environmental effects of urbanization. Urban trees sequester carbon through photosynthesis, decrease stormwater runoff by intercepting rainwater, reduce nutrient export through blue their root systems, and increase biodiversity by providing habitat and food to a variety of wildlife. Information on tree species composition and stand structure are necessary for understanding urban forest status and functions. It can be also used to assess the impacts of insect disturbance (e.g. Emerald ash borer, and Dutch elm disease), and help city foresters make management strategies.

Rapid advances in remote sensing technology enable efficient surveying of large areas of forest. Generally, remote sensing is to acquire information of objects on the earth through sensor instruments put on a platform such as an airplane or satellite. Different sensors are designed for a wide range of purposes and applications. For example, multi-spectral remote sensing has been widely used in global mapping of land cover and land use change. For the purpose of this project, two types of sensor (Lidar and AVIRIS hyperspectral sensors) are of particular interest (see Data section for more details).

OBJECTIVE AND DESCRIPTION OF DATA

The overarching goal of this research is to map the tree species composition for forests in certain areas in the U.S., using remote sensing data. The specific goal is to use *remote sensing data* to build predictive models for the proportion of abundance for a given species as measured by relative basal area (RBA) estimated from *field data*. The two types of data: *remote sensing data* and *field data* are described below.

Remote sensing data. The remote sensing data include two sources: (1) Hyperspectral imagery from NASA's AVIRIS sensor, and (2) Lidar data collected by the city of Madison. Hyperspectral data consist of reflectance measurements every 10 nm between 400 and 2500 nm, excluding wavelengths for which there are atmospheric absorption features that interfere with the remote sensing signal. This means that there are on the order of 182 "bands" of data.

Lidar data consist of measurements that come from laser pulses reflected from canopies, branches, or ground, and are summarized into a number of metrics of both height distribution from the lidar return and intensity distribution. Think of it this way: the hyperspectral information tells you the

reflectance of the canopy by sunlight, and the lidar data gives you information about the vertical structure of the canopy.

Different species have different reflectance characteristics and different vertical profiles that we can use to identify the composition of the species that occur in a location. The differences in the species may have to do with different leaf shapes (broad vs. needle leaves), or different chemistries, or different canopy shapes (think of how a conical fir tree is different from a more spherical oak). To that end, we have used the hyperspectral imagery and lidar data to map canopy chemistry and vertical structure, independent of species composition. Specifically, we have measures of foliar concentration of carbon, cellulose, fiber, lignin, leaf mass per area (LMA), nitrogen, as well as indices NDVI, PRI, NDWI and NDII capturing vegetation abundance, light use efficiency and water content, all derived from the hyperspectral images. In addition, we have biomass, basal area, average tree diameter, canopy height, crown length, and crown width, all derived from lidar.

Field data. For the field data, each observation is a measurement from a field plot sampled by expert forestry students. At each plot, the researchers have calculated the basal area (BA, trunk cross sectional area) of each species present at the plot. Composition can therefore be described two ways: basal area by species, or relative basal area by species. That is, for each species on a plot, we can divide its basal area by the total plot basal area to get an estimate of relative basal area. The relative basal area (RBA) by species is what we want to predict. At each location we are really only interested in correctly estimating species with RBA greater than about 10%. Species with less than 5-10% would just be considered low or rare, compared to other species. All estimates of species need to add to 100% (we cannot have more than 100% relative basal area). This gives us proportional estimates of composition by species. Later we can group these together to estimate forest types.

DATA FILE

The data set available includes:

- (1) Field data: plot ID, total basal area, basal area for 10 tree species, and basal area for the rest of species in the field plots;
- (2) Remote sensing data derived from lidar data: The measures of forest structure derived from lidar for all the listed field plots are shown in the columns AGB, BA, DBH, Ht, CL and CW;
- (3) Remote sensing data derived from hyperspectral data: The measures of canopy chemistry, LMA and indices derived from hyperspectral imagery for all the listed field plots are shown in the columns of C, Cellulose, Fiber, Lig, LMA, N, NDVI, PRI, NDWI and NDII, as well as standard deviations of these measures within the plot:

C_sd, Cellulose_sd, Fiber_sd, Lig_sd, LMA_sd, N_sd, NDVI_sd, PRI_sd, NDWI_sd, NDII_sd.

- (4) The datasheet with the “Data” tab lists field data and remote sensing data; and that with the “Description” tab gives more detailed description of each column of data.

QUESTIONS

- A. Conduct an exploratory data analysis to assess the relation between RBA and lidar/hyperspectral data. In particular, use a specific dimension reduction method and create visualizations for the relationship.
- B. Using hyperspectral data, construct a predictive model for the proportion of: (1) white oak (2) white pine and (3) ash. Provide statistical evaluation of prediction performance and statistical assessment of variable importance (for prediction) for your model. Remember that the summation of three predicted RBA for the three species should be always less than or equal to 1 (on any future data).
- C. Repeat the analysis of question B using the lidar data.
- D. An emerging idea is to combine both types of remote sensing data for the purpose of constructing a predictive model. Explore this possibility with each of the three species above (i.e., (1) white oak (2) white pine and (3) ash) and argue for or against this idea of combining hyperspectral data and lidar data.)
- E. What are the essential steps to take next for achieving the overarching goal of mapping tree species composition for forests in this study area, using remote sensing data? Provide your recommendations to the researchers and explain your reasoning.

Problem 2

Generate seven equally spaced points denoted by $(x_1, \dots, x_7)$, from the interval $[0,1]$. For each of these input points, generate seven responses $(y_1, \dots, y_7)$ using the function:

$$(1) \quad Y(x) = \exp(-1.4x) \cos\left(\frac{7\pi x}{2}\right).$$

- (1) Consider a Gaussian process model:

$$Y(x) = \mu + Z(x),$$

where μ is an unknown constant and $Z(x)$ is a Gaussian process with mean 0 and covariance function defined as $\text{cov}[Z(x+h), Z(x)] = \sigma^2 \phi(h; \theta)$. Assume the following form of the correlation function $\phi(h; \theta)$:

$$\phi(h; \theta) = \exp(-\theta h^2).$$

Write down the expressions for the maximum likelihood estimators (MLE) for parameters μ , θ , and σ^2 and calculate their values based on the generated seven pairs of observations (x_j, y_j) , $j = 1, \dots, 7$.

- (2) Write down the Best linear unbiased predictor (BLUP) of $Y(x_0)$ at $x = x_0$ and its predictive distribution after plugging in the MLEs of the parameters.
- (3) Now generate 1000 equally spaced points x_j^* , $j = 1, \dots, 1000$ over $(0,1)$. Generate the corresponding responses y using the same function (1).
- (4) Compare the MLE-EBLUP (the empirical BLUPs based on the MLEs) at input points x_j^* , $j = 1, \dots, 1000$ with the corresponding true values generated in (3) by plotting the predictions along with the true values.
- (5) Using the predictive distribution described in (2), add the 95% predictive confidence bands to the plot generated in (4).
- (6) Prove that the BLUP is an interpolator and demonstrate the interpolation property using your answers to (4) and (5).
- (7) Using the same training data (seven data points) used earlier to fit the Gaussian process model, fit a spline-based interpolating model to predict $Y(x)$ from x . Using the data generated in (3) as the testing data, compare the prediction performances of the Gaussian process model and the spline-based model and comment. Do you think there is any tangible advantage or disadvantage that one method has over the other, apart from their predictive performances? If so, mention it in a couple of sentences.